

BRAC UNIVERSITY
CSE331 : Automata and Computability
Summer 2025

Duration: 35 minutes

Quiz 1

Total: 20 marks

Name:

ID:

0.5 Points

Section:

0.5 Points

DFA & RE

$L_1 \rightarrow \{w \in \Sigma^* \mid w \text{ contains 1 in the second position}\}$, where $\Sigma = \{0, 1\}$

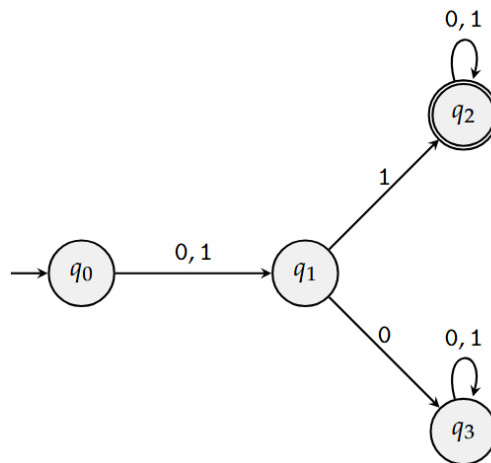
$L_2 \rightarrow \{w \in \Sigma^* \mid |w| \% 3 = 1\}$, where $\Sigma = \{0, 1\}$

$L_3 \rightarrow \{w \in \Sigma^* \mid \text{Every even position of } w \text{ is } 1\}$, where $\Sigma = \{0, 1\}$

$L_4 = L_1 \cap L_2$

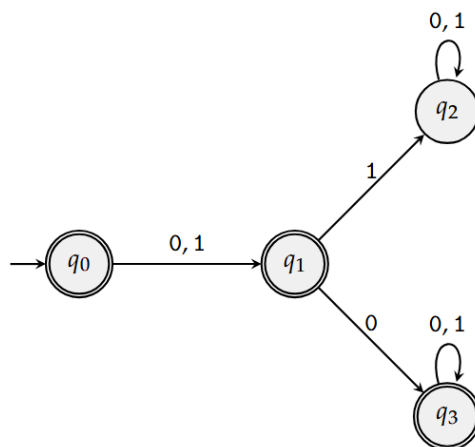
A. Draw the state diagram of a DFA for L_1 . (5 Points)

Answer:



B. Draw the state diagram of a DFA for $\overline{L_1}$. (2 Points)

Answer:



C. Write a Regular Expression for L_2 . (5 Points)

Answer: $(\Sigma\Sigma\Sigma)^*\Sigma$

D. Write a Regular Expression for L_2^* . (1 Point)

Answer: $((\Sigma\Sigma\Sigma)^*\Sigma)^*$

E. Write all four letter strings in L_4 . (2 Points)

Answer: 0100, 0101, 0110, 0111, 1100, 0101, 1110, 1111.

F. Write a Regular Expression for L_4 . (3 Points)

Answer: $\Sigma 1 \Sigma (\Sigma \Sigma \Sigma)^* \Sigma$

G. Is $L_3 \subset L_1$? (1 Point)

Answer: No.

Bonus

$L_1 \rightarrow \{w \in \Sigma^* \mid w \text{ contains 1 in the second position}\}, \text{ where } \Sigma = \{0, 1\}$

$L_2 \rightarrow \{w \in \Sigma^* \mid |w| = 3\}, \text{ where } \Sigma = \{0, 1\}$

Write a Regular Expression for the following regular language: (2 Points)

$L(M) \rightarrow \overline{L_1 \cap L_2}$

Answer:

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Quiz 2

Total: 20 marks

Name:

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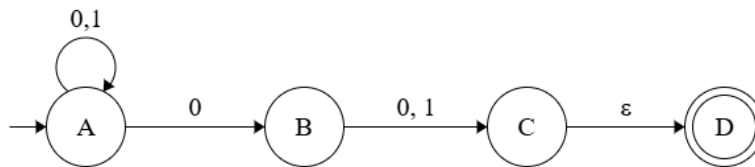
0.5 Points

Section:

0.5 Points

NFA→DFA

1. Consider the following NFA:



A. If you convert the given NFA into an equivalent DFA using the subset construction method, what is the maximum number of rejecting states the DFA can have? (1 Mark)

Answer: 8

B. Write the ϵ -closure of state C in the given NFA. (1 Mark)

Answer: {C, D}

C. What is $\delta(\{A, B\}, 1)$ in the given NFA? (1 Mark)

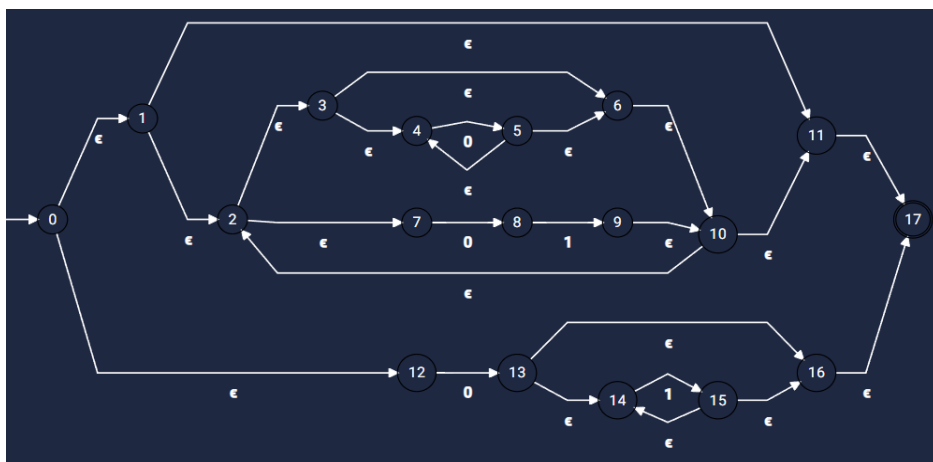
Answer: {A, C}

RE→NFA

2. Convert the following RE into an equivalent NFA using the Thompson's construction: (7 Marks)

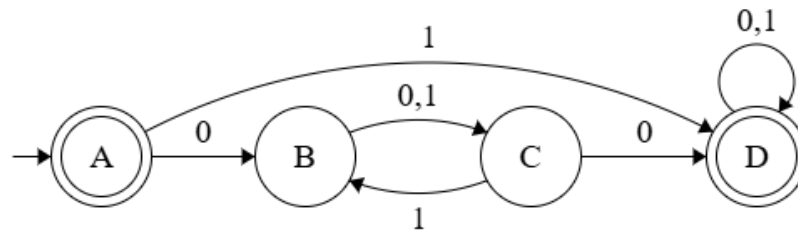
$(0^* \cup 01)^* \cup 01^*$

Solution:



FA→RE

3. Convert the following DFA into an equivalent RE using the state elimination method(Eliminate the states in the following order: D, C, B, A): (10 Marks)



Solution: $1\Sigma^* \mid 0(\Sigma 1)^*\Sigma 0\Sigma^* \mid \varepsilon$

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Quiz 3

Total: 20 marks

Name:

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0.5 Points

Section:

0.5 Points

Pumping Lemma

Use the pumping lemma to prove that the following language is not regular: (8 points)

$L(M) \rightarrow \{w \in \{0,1\}^* \mid w \text{ contains more numbers of 0s than 1s}\}.$

Solution:

$$w = 1^p 0^{p+1}$$

For $i = 2$,

$$xy^2z = 1^a 1^{2b} 1^{p-a-b} 0^{p+1}$$

$$\Rightarrow b < 1$$

CFG

$L_1 \rightarrow \{w \in \{0, 1\}^* \mid w \text{ contains equal numbers of 0s and 1s}\}$

$$R_1 = 0\Sigma^*1 \mid 1\Sigma^*0$$

$$L_2 = L_1 \cap L(R_1)$$

A. Write a CFG for L_1 . (3 Marks)

Answer:

$$S \rightarrow 0S1 \mid 1S0 \mid SS \mid \varepsilon$$

B. Convert R_1 into an equivalent CFG. (3 Marks)

Answer:

$S \rightarrow A \mid B$

$A \rightarrow 0X1$

$B \rightarrow 1X0$

$X \rightarrow 0X \mid 1X \mid \epsilon$

C. Write all four-letter strings in L_2 . [Note: $L(R_1)$ denotes the language generated by the regular expression R_1] (2 Marks)

Answer: 0011, 0101, 1010, 1100.

D. Write a CFG for L_2 . (3 Marks)

Answer:

$S \rightarrow 0A1 \mid 1A0$

$A \rightarrow 0A1 \mid 1A0 \mid AA \mid \epsilon$

Bonus

Use the pumping lemma to prove that the following language is not regular: (2 points)

$L(M) \rightarrow \{w_1w_2 \mid w_1 \text{ is a substring of } w_2 \text{ and } |w_1| \neq |w_2|\}$, where $\Sigma = \{0, 1\}$.

Solution:

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Quiz 4

Total: 20 marks

Name: _____ **ID:** _____ **0.5 Points**

Section: _____ **0.5 Points**

Ambiguity

Consider the following Context-Free Grammar, **G**:

$A \rightarrow 0B \mid 1A \mid 1C \mid \epsilon$

$B \rightarrow 0A \mid 1B$

$C \rightarrow 0C0 \mid 1C1 \mid \epsilon$

a. Write a leftmost derivation for the string 10011 using **G**. (2 points)

Answer: $A \Rightarrow 1A \Rightarrow 10B \Rightarrow 100A \Rightarrow 1001A \Rightarrow 10011A \Rightarrow 10011$

b. Show that **G** is ambiguous by writing another leftmost derivation of the given string. (2 points)

Answer: $A \Rightarrow 1A \Rightarrow 10B \Rightarrow 100A \Rightarrow 1001A \Rightarrow 10011C \Rightarrow 10011$

c. Write a three-letter string that has only one parse tree using **G**. (1 point)

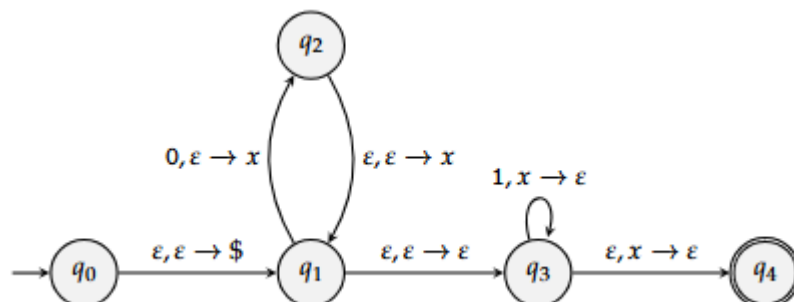
Answer: 010

PDA

1. Draw the state diagram of a PDA for the following CFL: (6 points)

$L(M) \rightarrow \{w = 0^n 1^m \mid n, m \geq 0 \text{ and } 2n > m\}, \text{ where } \Sigma = \{0, 1\}$

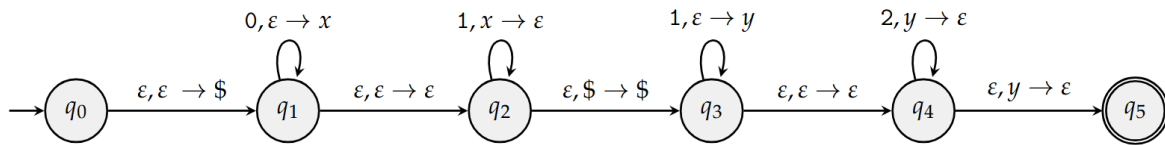
Solution:



2. Draw the state diagram of a PDA for the following CFL: (8 points)

$L(M) \rightarrow \{w = 0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and } j > i + k\}, \text{ where } \Sigma = \{0, 1, 2\}$

Solution:



Bonus

1. Draw the state diagram of a PDA for the following CFL: (2 points)

$L(M) \rightarrow \{w = 0^n 1^m \mid n, m \geq 0 \text{ and } m \leq n \leq 2m\}, \text{ where } \Sigma = \{0, 1\}$

Solution:

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Quiz 5

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Name:

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0.5 Points

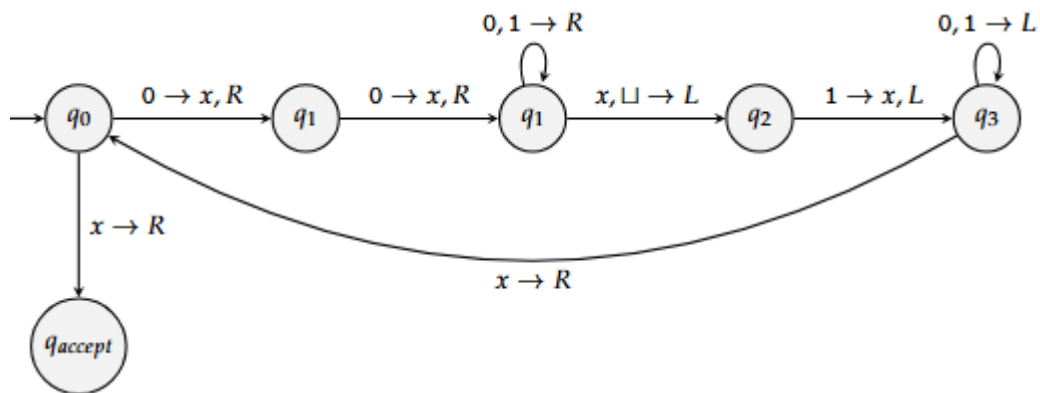
Section:

0.5 Points

Turing Machines

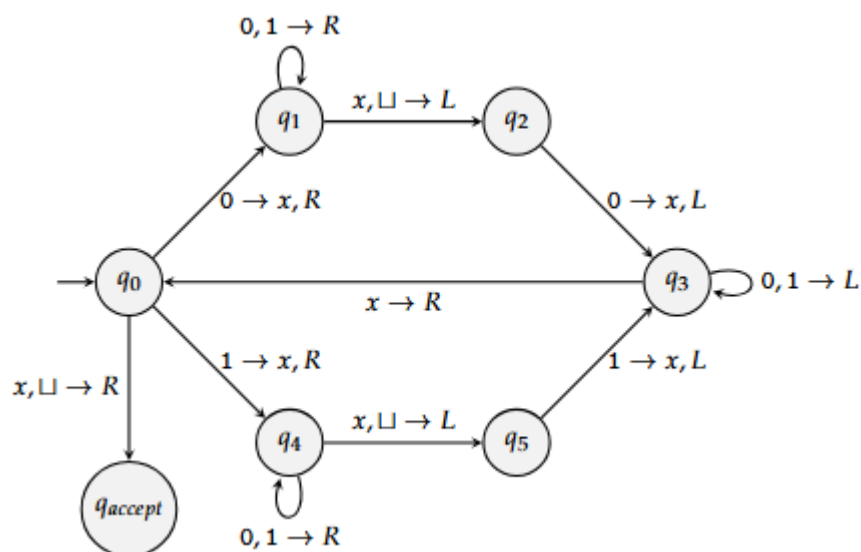
1. Let, $L_1 \rightarrow \{w = 0^n 1^m \mid n, m > 0 \text{ and } n = 2m\}$, where $\Sigma = \{0, 1\}$. Draw the state diagram of a TM that decides L_1 . (5 points)

Solution:



2. Let, $L_2 \rightarrow \{w \in \Sigma^* \mid w \text{ is an even length palindrome}\}$, where $\Sigma = \{0, 1\}$. Draw the state diagram of a TM that decides L_2 . (6 points)

Solution:



3. Write the current state and current head position of a TM in a configuration 010q₆10. (2 points)
Answer: q₆, 4th symbol.

5. Let, $A_{CORE} \rightarrow \{\langle R, w \rangle \mid R \text{ is a regular expression that does not generate the string } w\}$.

Prove that A_{CORE} is decidable. (6 points)

Solution: M = “On input $\langle R, w \rangle$:

1. Convert regular expression R to an equivalent NFA N using Thompson's construction method.
2. Convert NFA N to an equivalent DFA D using subset construction method.
3. Simulate D on input w.
4. If D accepts, reject; otherwise accept.”

Since the simulation always ends after $|w|$ steps, M is a decider.

Bonus

1. Let, $A_{TM} \rightarrow \{\langle M, w \rangle \mid M \text{ is a TM that accepts } w\}$. Prove that A_{TM} is undecidable. (2 points)

Solution: